

Link to code: <https://github.com/luca-miniati/ds4400/blob/main/hw3/hw3.ipynb>

Solution 1

1. Implemented in code
2. The top 10 most predictive features with their coefficients are:

Rank	Feature	Coefficient
1	word_freq_george	-3.746023
2	char_freq_\$	3.532032
3	word_freq_000	2.023771
4	word_freq_hp	-2.023331
5	word_freq_remove	1.972562
6	word_freq_meeting	-1.858157
7	word_freq_edu	-1.627047
8	word_freq_project	-1.330245
9	word_freq_lab	-1.316705
10	char_freq_#	1.305303

3. As T increased, precision increased and recall decreased. Accuracy peaked at $T = 0.5$.

Solution 2

1. The sklearn model performed way better than my implementation. Probably, scaling the data and training for longer would help.

Solution 3

1. Implemented in code
2. The best model was Logistic Regression. Second-best was LDA, and KNN performed worst.
3. Implemented in code
4. To make the manual ROC curve more like the sklearn one, I can use more thresholds.

Solution 4

1. Implemented in code
2. Implemented in code
3. For both values of k , Logistic Regression had lower error.